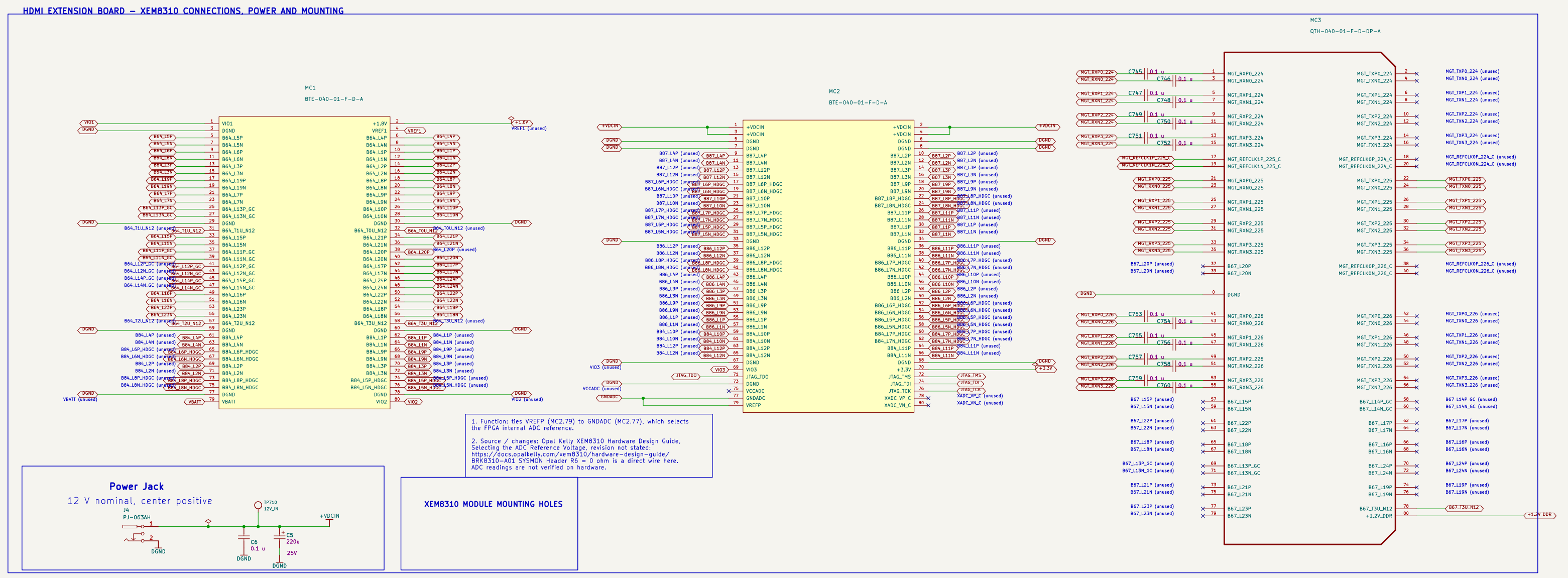
HDMI EXTENSION BOARD – XEM8310 CONNECTIONS, POWER AND MOUNTING



1. Function: ties VREFP (MC2.79) to GNDADC (MC2.77), which selects the FPGA internal ADC reference.

2. Source / changes: Opal Kelly XEM8310 Hardware Design Guide, Selecting the ADC Reference Voltage, revision not stated: https://docs.opalkelly.com/xem8310/hardware-design-guide/ BRK8310-A01 SYSMON Header R6 = 0 ohm is a direct wire here. ADC readings are not verified on hardware.

Power Jack

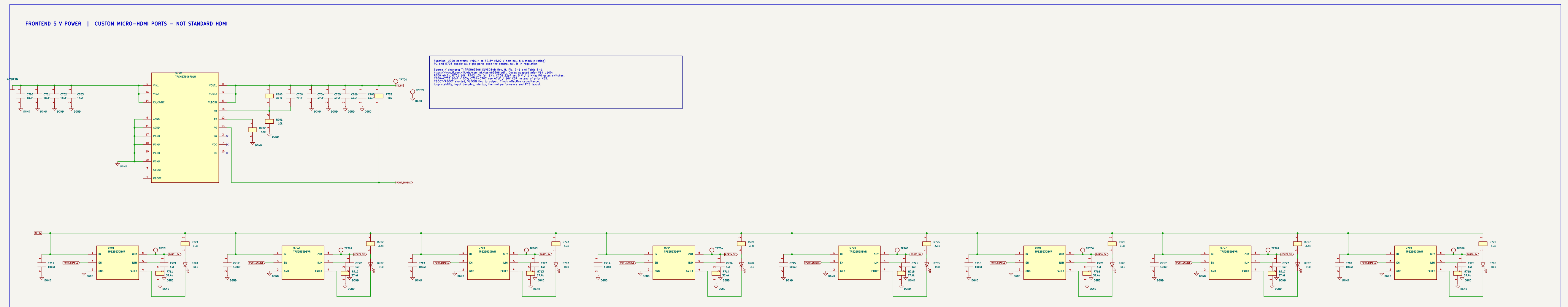
12 V nominal, center positive

XEM8310 MODULE MOUNTING HOLES

Function: XEM8310 connectors provide the board power, FPGA I/O, JTAG and GT connections; J4 is the 12 V input; the module is held by its connectors and two M3 standoffs (holes MH2/MH3 on the board, restored in V36; MH1/MH4 are routed over).

Source / changes: Opal Kelly BRK8310-A01 p.1 (2022-05-19): https://pins.opalkelly.com/downloads/442/download?category_id=3 ; Codex assigned the custom HDMI and x4 nets. VIO1 target is 1.8 V; program and verify it. V24: J4 changed from CUI PJ-102AH (5 A) to Same Sky PJ-063AH (8 A, 24 V, same 2.0 mm pin and 5.5 mm plug, no switch pin; unused PJ-102AH pin 3 dropped): https://www.sameskydevices.com/product/resource/pj-063ah.pdf ; footprint = KiCad library BarrelJack_CUI_PJ-063AH_Horizontal, pin 1 at the old pin 1. Mating plug and 12 V adapter must also be rated above 5 A; footprint not re-measured against the drawing. V29 (Howard, 4 Nov 2026): BRK8310 standoff holes MH3/MH4 beside MC3 removed for routing space; the XEM8310 is now held by standoffs at MH1/MH2 only and by its connectors at the MC3 end (mechanical retention not verified). V45 (Howard, 7 Nov 2026): MH1/MH2 removed too; the XEM8310 is held only by MC1/MC2/MC3 (retention not verified).
V33 (Howard 6 Nov 2026), GPIO command (GT command dropped, Howard 6 Nov 2026): per port 1–8, HDMI 9/11 = command with SYNC folded in on L5, L3, L7, L15, L4, L2, L9, L21 and 6/8 = 32 MHz clock on L6, L19, L13_GC, L11_GC, L1, L8, L10, L24 (LVDS outputs, VIO1 1.8 V; pairs chosen so each port's clock and command reach MC1 side by side). MC3 quad 224/226 GTY TX is unused (quad 225 TX = PCIe); 12/14: see V34/V36 below. The recording RX pairs (3/5) are AC-coupled by C745–C760 100 nF at MC3 instead of on the Frontend (Howard 5 Nov 2026); the XEM8310 pin list marks MC3 GT RX as not AC-coupled on the module and AMD UG482 v1.9 pp.235–236 (https://docs.amd.com/v/u/en-US/ug482_7Series_GTP_Transceivers; GTY receive side: UG578 v1.4, https://docs.amd.com/v/u/en-US/ug578-ultrascale-gty-transceivers, page not checked) only requires the 100 nF somewhere between transmitter and receiver. Update the XDC to match.
V34 (Howard 7 Nov 2026): separate SYNC on HDMI 12/14 for ports 3, 4, 5, 6 only (port 3 added in V36), on the free bank 64 pairs L16 (port 3), L23 (port 4), L22 (port 5) and L18 (port 6), LVDS outputs at VIO1 1.8 V. SYNC stays folded into the 9/11 command on every port; the extra pair lets SYNC be split later on this subset. Ports 1, 2, 7 and 8 have no free route to MC1 (12/14 clamped only). L12_GC and L14_GC stay unused. The Frontend must take 12/14 on LVDS-capable pins for these pairs to be usable.

FRONTEND 5 V POWER | CUSTOM MICRO-HDMI PORTS – NOT STANDARD HDMI

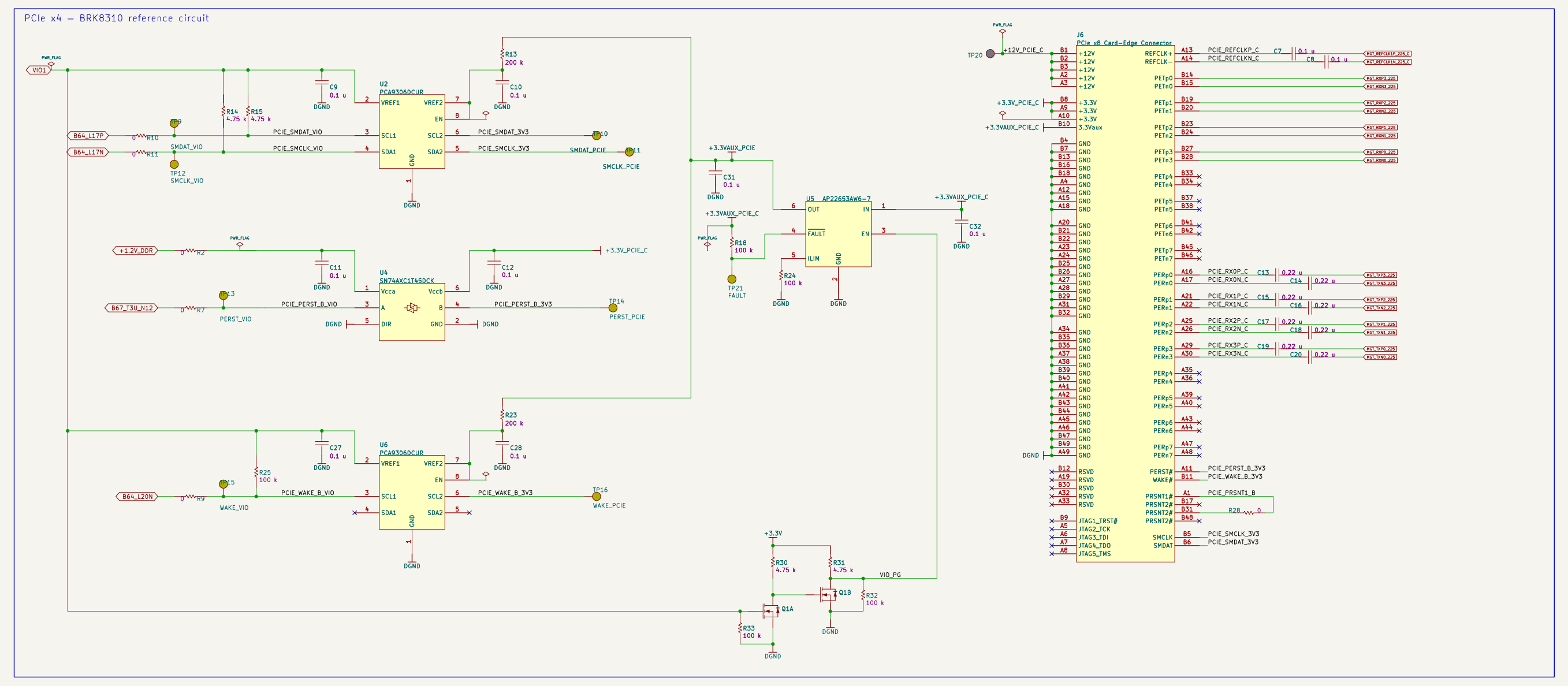


Function: U700 converts +VDCIN to FE_5V (5.02 V nominal, 6 A module rating).
PG and R703 enable all eight ports once the central rail is in regulation.

Source / changes: TI TPSM63606 SLVSGB4B Rev. B, Fig. 9–1 and Table 8–1,
https://www.ti.com/lit/ds/symlink/tpsm63606.pdf . Codex adapted prior V14 U100:
R700 40.2k, R701 10k, R702 13k (all 1%), C708 22pF set 5 V / 1 MHz; PG gates switches.
C700–C703 10uF / 50V; C704–C707 use 47uF / 16V X5R instead of prior X6S.
CBOOT/RBOOT shorted, VLDOIN tied to output. Check effective capacitance,
loop stability, input damping, startup, thermal performance and PCB layout.

Function: U701–U708 individually switch FE_5V to J610–J617 pin 19; red LEDs indicate each switch FAULT. TP701–TP708 access the outputs. All ports start together; no staggering.
Source / changes: TI TPS2553 SLVS841G Rev. G, pins / sections 8–9 and Table 9–2: https://www.ti.com/lit/ds/symlink/tps2553.pdf . Codex adapted prior V14; EN now follows U700 PG;
R711–R718 = 37.4k 1% (630–771 mA limit); C711–C718 are IN bypass; V31: C721–C728 1 uF (Samsung CL10A105KB8NNNC, LCSC C15849) added on each OUT / PORTn_5V per SLVS841G sec. 9.2.1.2.4 p. 21 and the p. 1 ESD rating with external capacitance; value chosen by Claude, not given by TI; R721–R728 / D701–D708 added as FAULT lights. Each OUT goes to a separate pin-19 net.
Cable current and return rating, hot-plug / ESD, backfeed, 8-port inrush and thermal behavior are not verified. Existing 3.3 V-only frontends must NOT be connected to these 5 V ports.

PCIe x4 – BRK8310 reference circuit

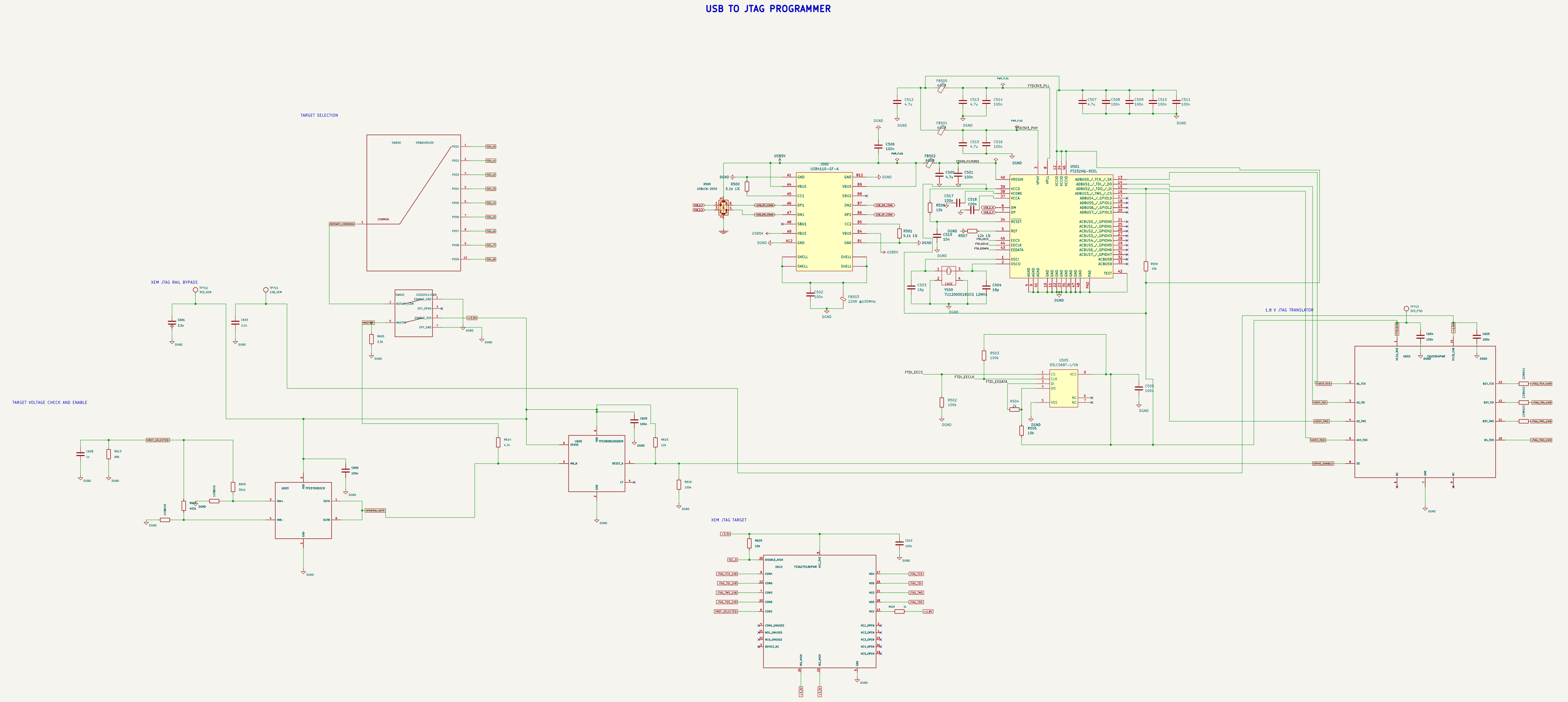


Function: Connects PCIe x4 data and 100 MHz reference clock, with reset, SMBus and WAKE#. Host slot 12 V is isolated from the board 12 V input.

Source / changes: Opal Kelly BRK8310-A01 schematic, p. 2, 2022–05–19: https://pins.opalkelly.com/downloads/442/download?category_id=3
Adapted from x8 to x4: fit R28 = 0 ohm; omit R29 and unused lanes 4-7 and C21-C26/C29-C30. R2/R7 and R9-R11 retained.
Unused options R1/R8, R16/R17 and R27 are omitted.
Host +12 V goes only to TP20. Host 3.3 V and AUX power the original sideband circuitry; they do not feed the main 12 V input.
PCA9306 Rev O, Fig. 8–1, section 10: https://www.ti.com/lit/ds/symlink/pca9306.pdf ; SN74AXC1T45 Rev E: https://www.ti.com/lit/ds/symlink/sn74axc1t45.pdf
AP22653 DS41186 Rev 5–2, sections 7-9: https://www.diodes.com/datasheet/download/AP22652_AP22653_AP22652A_AP22653A.pdf
Q1 Si1922EDH dual FET, product datasheet: https://www.vishay.com/en/product/67192/
V30 (Howard, 4 Nov 2026): TX AC–coupling caps C13–C20 changed from BRK8310's 0.1 µF (Kemet C0402C104K8PACTU) to 0.22 µF (Samsung CL05B224KO5NNNC, LCSC C16772, 0402 X7R 16 V); same footprint and routing.
Reason: PCIe Gen3 TX coupling range 176-265 nF (commonly cited PCIe Base Spec value, also in AMD UG583; the PCI-SIG document itself was not read); 220 nF also covers Gen1/2. C7/C8 (REFCLK) unchanged.
Working target: VIO1 = 1.8 V (Howard, 3 Nov 2026; needed for PCA9306 >= 1.2 V and Bank 64 LVDS); set in XEM before testing and verify all Bank 64 I/O. Factory default 1.0 V is below PCA9306 range: https://docs.opalkelly.com/xem8310/device-settings/
ERC power flags at U2/U6 VREF2 represent their R13/R23 200k resistor-fed bias, not independent regulators. Confirm bias and startup.
PCIe/HDMI combined Vivado design, clock, VIO1 choice, host sequencing and signal integrity remain to be verified.

USB-C JTAG PROGRAMMER, SELECTION AND FRONTEND PORTS

USB TO JTAG PROGRAMMER



Function: J500 and U501 convert USB–C to JTAG; U505 stores the programmer settings.
U602 translates the JTAG lines to 1.8 V. SW600 selects the XEM or one of eight frontends.
U603 checks the selected target voltage; U605 enables the translator after a delay.
C601/C603 bypass the XEM rails; U610 selects the XEM path. Turn OFF before selecting.

Source / changes: USB–C input: Codex adaptation of BRK8305 A03,
25Oct2023, JTAG sheet8/PDFp4: https://pins.opalkelly.com/downloads/583/download?category_id=3
J500=USB4110–GF–A; D500=USBLC6–2SC6. C506
changed 10nF to 100nF. USB shell uses C502/FB503;
shared DGND. Connector fit and USB/ESD layout unverified.
ST USBLC6–2 Rev7 Fig15: https://www.st.com/resource/en/datasheet/usblc6–2.pdf

FT232H and EEPROM: Codex adaptation of BRK8305 A03,
25Oct2023, JTAG sheet8/PDFp4: https://pins.opalkelly.com/downloads/583/download?category_id=3
FT232HQ pins13/14/15/16=TCK/TDI/TDO/TMS; ADBUS4–7
and ACBUS NC as BRK8305. Former R508 removed.
U505=93LC56BT–I/SN (SOIC) replaces CFG001/MSOP;
Y500=SJK replaces Abracon; FB500-502=Murata BLM18AG601SN1D.
Y500 specifies 18pF load; 18pF C503/C504 give ~9pF + stray.
Confirm crystal load, frequency, drive and startup after layout.
Y500: https://xonstorage.z8.web.core.windows.net/pdf/sjk_7u12000e18ucg_xonjuly20_20_link.pdf
Bead current and voltage drop need checks.
Microchip DS21794F Table3–1: https://ww1.microchip.com/downloads/en/DeviceDoc/21794F.pdf
FT232H DS v2.2 sec3,6: https://ftdichip.com/wp–content/uploads/2024/09/DS_FT232H.pdf
AMD program_ftdi EEPROM write/readback and Vivado
enumeration must be tested. VCK190 match not established.
https://docs.amd.com/r/2024.1–English/ug908–vivado–programming–debugging/Programming–FTDI–Devices–for–Vivado–Hardware–Manager–Support

Translator: Codex adaptation; TI TXU0304 Rev A sec.9: https://www.ti.com/lit/ds/symlink/txu0304.pdf
VCCA = FTDI3V3; VCCB = XEM +1.8 V (MC1.2); R601-R603 = 22R provisional.
R604 pull–up removed; IC inputs have weak pulldowns. Check rail sequencing, unpowered–target
injection, timing, cable loading and FTDI/Vivado operation.

Rail bypass: Codex adaptation: C601/C603 retained from the removed regulators.
XEM8310 MC2.70 = +3.3 V and MC1.2 = +1.8 V; rail budget and startup must be verified.
Opal Kelly XEM8310 pin list: https://pins.opalkelly.com/pin_list/XEM8310

Rotary selector: Codex direct–to–enable wiring; nine 10k pulls replace decoder logic.
ALPS SRBQ 1P9T non–shorting; confirm stock, contact sequence and footprint.
https://tech.alpsalpine.com/e/products/detail/SRBQ490100/
Bounce and switch faults are not proven.

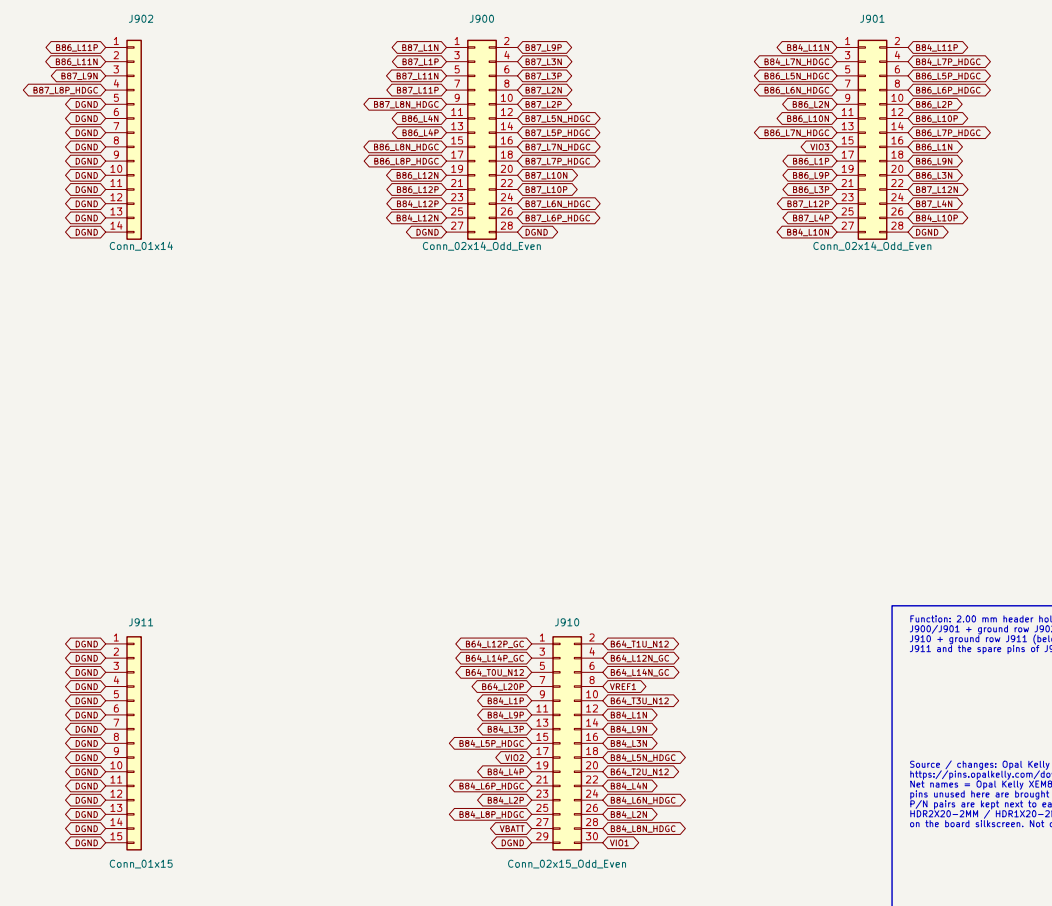
OFF/ENABLE: Codex dual–pole arrangement; R600 biases MASTER low.
C&K JS datasheet p.9; verify contact positions and footprint:
https://www.littelfuse.com/assetdocs/littelfuse–c–k–slide–js–series–datasheet?assetguid=aba42b08–0d2c–423b–813d–a2faa5a3bb14

Voltage window: Codex circuit; TI TPS3700 Rev G sec.7.3: https://www.ti.com/lit/ds/symlink/tps3700.pdf
XEM 3.3 V supplies U603. Selected VREF uses XEM R629 or HDMI pin 1.
R605-R608=301k/100k and 402k/100k; R613/C608 discharge/filter.
Nominal window 1.604-2.008 V; verify leakage, tolerance and startup.

Delay: Codex circuit; TI TPS3808 Rev N secs.4, 7: https://www.ti.com/lit/ds/symlink/tps3808.pdf
U605 changed G33 to G30: 2.79 V nominal threshold increases 3.3 V rail margin.
CT open: nominal 20 ms; R615=12k now pulls to XEM 3.3 V, R616=100k pulls down.
Startup and brownout remain unverified; keep OFF during power–up.
JLC C189211 listing conflicts with TI G30 threshold; verify MPN/marking.

XEM target switch: Codex circuit using TI TS3A27518E Rev F Fig.5–2, Table 8–1:
https://www.ti.com/lit/ds/symlink/ts3a27518e.pdf
IN1/IN2 high select NO; active–low EN from rotary; 10k pullup.
Channels (Howard 1 Nov 2026, for straight PCB routing):
TCK=4, TDI=6, TMS=3, TDO=5, VREF=2; channel 1 unused.
NO2–NO6: ±6 kV IEC contact rating; not a 5 V DC clamp.
Selection, leakage, power–off and SI need checks.
XEM JTAG pins: https://docs.opalkelly.com/xem8310/expansion–connectors/

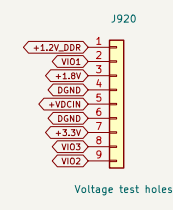
XEM8310 unused–pin header holes (V45; bare holes since V46)



Function: 2.00 mm header holes for every XEM8310 MC1/MC2 pin this board does not use (Chong via Howard, 7 Nov 2026).
J900/J901 + ground row J902 (top–right corner): MC2 banks 84/86/87 and VIO3.
J910 + ground row J911 (below MC1): MC1 banks 64/84 with VIO1, VIO2, VREF1, VBATT. J902 pins 1–4 carry I/O and 5–14 are DGND;
J911 and the spare pins of J900 (27, 28), J901 (28) and J910 (29) are DGND.

Source / changes: Opal Kelly BRK8310 breakout, 2.00 mm headers MC4–MC11 (BRK8310–A01 p.1, 2022–05–19:
https://pins.opalkelly.com/downloads/442/download?category_id=3 ; native import ~/kicad_work/brk_ref/BRK8310.PcbDoc).
Net names = Opal Kelly XEM8310 pin list (https://pins.opalkelly.com/pin_list/XEM8310). Changed from BRK: only the
pins unused here are brought out; Codex chose the header sizes (2x14, 2x15), places and pin order from the board routing;
P/N pairs are kept next to each other where the routing allowed. V46 (Howard, 7 Nov 2026): bare header holes like BRK8310
HDR2X20–2MM / HDR1X20–2MM (0.8 mm drill, 1.3 mm pad), no part fitted (not in BOM or placement); every hole is named
on the board silkscreen. Not checked: signal integrity on these lines.

Voltage test holes (V35)

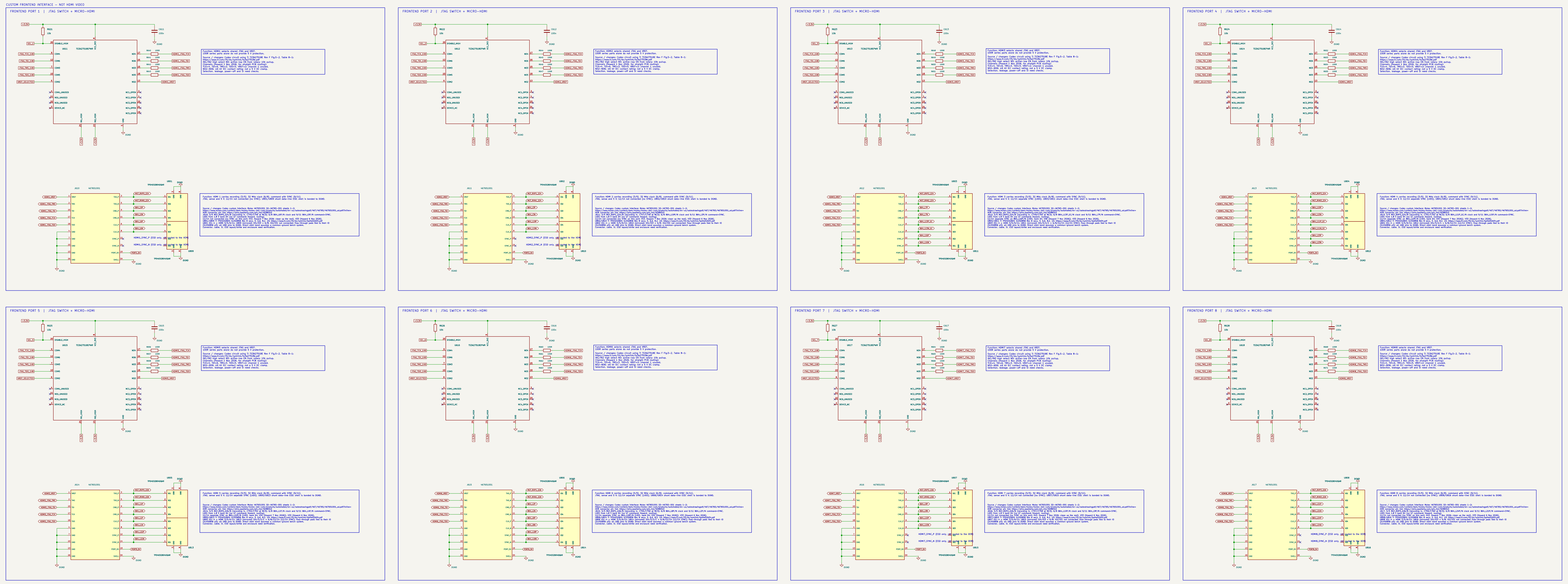


Function: nine bare test holes for probing the board rails (Chong via Howard, 8 Nov 2026: "is it difficult to fit these voltage header holes as well?"). Holes left to right: +1.2V_DDR, VIO1, +1.8V, DGND, +VDCIN (12 V in), DGND, +3.3V, VIO3, VIO2.
Row below MC1, next to J910/J911; each hole is named on the top and bottom silkscreen. No part fitted (not in BOM or placement).

Source / changes: Opal Kelly BRK8310–A01 schematic p.1 (2022–05–19), test points TP1–TP4, TP7, TP8, TP17–TP19 (footprint Aligni:100, 1.651 mm drill, 2.54 mm pad): https://pins.opalkelly.com/downloads/442/download?category_id=3 ; layout = native import of BRK8310.PcbDoc (9–hole column at 6.18 mm pitch: DGND, +VDCIN, +1.8V, +3.3V, +1.2V_DDR, VIO1–3, DGND; extra DGND points TP5/TP6 not copied).
Changed by Claude: one 1x9 footprint J920 at 3.25 mm pitch in a row; order changed so 12 V sits only beside DGND, VIO1 only beside <=1.8 V rails, and every feed routes without crossings. 0.2 mm feeds from MC1.2, MC2.1/MC2.69, R2 (+1.2V_DDR), the L4 +3.3V plane, J910.17 and the VIO1 run. Measure only. Not checked: probe/clip fit at 3.25 mm pitch. No series resistors (BRK has none): a probe slip between neighbours (e.g. +3.3V/VIO3/VIO2, +1.2V_DDR/VIO1) still connects two rails.

CUSTOM FRONTEND INTERFACE – NOT HDMI VIDEO

FRONTEND PORT 1 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 2 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 3 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 4 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 5 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 6 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 7 | JTAG SWITCH + MICRO–HDMI
FRONTEND PORT 8 | JTAG SWITCH + MICRO–HDMI



Function: HDMI1 selects shared JTAG and VREF.
100R series parts alone do not provide 5 V protection.

Source / changes: Codex circuit using TI TS3A27518E Rev F Fig.5–2, Table 8–1:
https://www.ti.com/lit/ds/symlink/ts3a27518e.pdf
IN1/IN2 high select NO; active–low EN from rotary; 10k pullup.
Channels (Howard 1 Nov 2026, for straight PCB routing):
TCK=4, TDI=6, TMS=3, TDO=5, VREF=2; channel 1 unused.
NO2–NO6: ±6 kV IEC contact rating; not a 5 V DC clamp.
Selection, leakage, power–off and SI need checks.

Function: HDMI 1 carries recording (3/5), 32 MHz clock (6/8), command with SYNC (9/11),
JTAG, sense and 5 V; 12/14 not connected (no SYNC). U801/U809 shunt data–line ESD; shell is bonded to DGND.

Source / changes: Codex custom interface; Molex 467651001 SD–46765–001 sheets 1–2:
https://www.molex.com/content/dam/molex/molex–dot–com/products/automated/en–us/salesdrawingpdf/467/46765/467651001_sd.pdf?inline=
XEM connector pin list: https://pins.opalkelly.com/pin_list/XEM8310
J610: 3/5 MGT_RXP0_224/N (recording in, C745/C746 at MC3); 6/8 B64_L6P/N clock and 9/11 B64_L5P/N command+SYNC,
LVDS from 1.8 V bank 64 (no GT command: Howard, routing).
12/14 not connected (no SYNC on this port; V43, Howard 7 Nov 2026: clean up the rest). V33 (Howard 6 Nov 2026).
U801/U809 TI TPD4E02B04 SLVSD85B Rev B secs. 5, 6.6, 8.2, 10: https://www.ti.com/lit/ds/symlink/tpd4e02b04.pdf
U801 IO1–4 = HDMI 3/5/9/11; U809 (mirrored) IO4/IO3 = 6/8; IO2/IO1 not connected; flow–through pads tied to their IO
(SLVSD85B p3); all GND pins to DGND. Direct shell bond assumes a common–ground bench system.
Connector, cable, SI, ESD layout/strike and enclosure need verification.

V37 (10 Nov 2026) = V36 + L6 ground pour under U700 (heat spreading for the 5 V module)

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Title: HDMI Extension Board – XEM8310 and USB–C JTAG

Size: User	Date: 2026–11–10	**Rev: V37**
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